

## 0/24 Questions Answered

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## hw8-diff-means-regression-dog-lifespan

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## Q1

1 Point

**Note: There are three options for homework 8, that differ by data topic, and you only need to complete one of the three assignments.**

Dog breeds can vary dramatically in characteristics such as size and lifespan. da Silva and Cross (2023) recorded the adult lifespan (in years) and adult body weight (in pounds) of 73 different dog breeds. We will use this data to investigate differences in mean lifespan based on weight.

|                  | Lifespan $\bar{x}$ | Lifespan $s$ | $n$ |
|------------------|--------------------|--------------|-----|
| Small dog breeds | 12.1448            | 1.4720       | 33  |
| Large dog breeds | 10.2125            | 2.0131       | 40  |

We will consider a large dog to be a dog over 50 pounds, and a small dog to be a dog under 50 pounds. We want to investigate if there is a difference in mean lifespan between small dog breeds and large dog breeds. Let

- $\mu_1$  = mean lifespan of small dog breeds
- $\mu_2$  = mean lifespan of large dog breeds

## Q1.1

1 Point

Is this an example of paired or unpaired data?

Unpaired

Paired

Save Answer

**Q2****7 Points**

We will conduct a hypothesis test to investigate if there is evidence of a difference in mean lifespan between small and large dog breeds, at a significance level of 0.05.

**Q2.1****1 Point**

Select the appropriate hypotheses.

$$H_0 : \mu_1 - \mu_2 \neq 0$$

$$H_1 : \mu_1 - \mu_2 = 0$$

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_1 : \mu_1 - \mu_2 \neq 0$$

$$H_0 : \mu_2 - \mu_1 \neq 0$$

$$H_1 : \mu_2 - \mu_1 = 0$$

$$H_0 : \mu_2 - \mu_1 = 0$$

$$H_1 : \mu_2 - \mu_1 \neq 0$$

Save Answer

**Q2.2****1 Point**

What is the point estimate for the difference in population mean lifespan between small and large dog breeds? Provide your final answer, rounding to four decimal places (make sure to keep the order of groups in mind).

Save Answer

**Q2.3****1 Point**

What is the standard error of the point estimate? Round to four decimal places. Upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q2.4****1 Point**

Our next step is to calculate the p-value for our test. We know the difference of means is approximately t-distributed, but what are the degrees of freedom of this t-distribution?

$$df = 33 - 1 = 32$$

$$df = 40 - 1 = 39$$

$$df = 33 + 40 = 73$$

**Save Answer**

**Q2.5****1 Point**

Calculate the p-value and round to 4 significant digits (recall you can enter  $2.354 \times 10^{-6}$  as `2.354e-6`). Upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q2.6****2 Points**

You should have found a very small p-value, much smaller than the significance level. Specify whether we reject the null hypothesis in support of the alternative hypothesis or if we fail to reject the null hypothesis, and then provide a conclusion in context based on the p-value.

**Save Answer**

**Q3****3 Points**

Now we are not just interested in investigating if there is a difference, we actually want to estimate the difference in means. Let's calculate a 95% confidence interval to express our uncertainty in our point estimate.

**Q3.1****1 Point**

What is the critical value for a 95% confidence interval for a mean? Round to four decimal places.

[Save Answer](#)**Q3.2****1 Point**

What is the lower bound of the 95% confidence interval for the difference of means? Round to four decimal places. Upload a picture of your work.

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

**Q3.3****1 Point**

What is the upper bound of the 95% confidence interval for the difference of means? Round to four decimal places. Upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer**



**Q4****5 Points**

In this homework we conducted a hypothesis test to test if there was evidence of a difference between the two population means. Then we calculated a confidence interval to estimate the difference between the population means.

What if instead we just want to study the relationship between lifespan and average weight for dog breeds, where weight is a continuous variable (instead of discretized into large and small)?

Let's review simple linear regression:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Identify each component of the equation by selecting its definition.

**Q4.1****1 Point**

$x_i$  is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

**Q4.2****1 Point** $y_i$  is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

**Q4.3****1 Point** $\epsilon_i$  is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

**Q4.4****1 Point** $\beta_1$  is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

**Q4.5****1 Point** $\beta_0$  is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

**Q5****4 Points**

In our case, the variable we want to learn about is lifespan and weight is the explanatory variable. We wanted to understand if there is a linear relationship between lifespan and weight, at a significance level of 0.05.

**Q5.1****1 Point**

What is the correct set of hypotheses to test if the mean lifespan linearly varies by dog breed size?

$$H_0 : \beta_0 = 0$$

$$H_a : \beta_0 \neq 0$$

$$H_0 : \beta_0 \neq 0$$

$$H_a : \beta_0 = 0$$

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

$$H_0 : \beta_1 \neq 0$$

$$H_a : \beta_1 = 0$$

Save Answer

## Q5.2

1 Point

Below we fit the simple linear regression model and show the output.

```
> dog_model <- lm(lifespan ~ weight_lb, data = dogs)
> broom::tidy(dog_model)
# A tibble: 2 x 5
  term          estimate std.error statistic    p.value
<chr>          <dbl>      <dbl>      <dbl>    <dbl>
1 (Intercept)    13.4        0.317      42.2 5.05e-52
2 weight_lb    -0.0422      0.00495    -8.54 1.68e-12
```

What is the p-value for our test? Write your answer with three significant digits.

Save Answer

## Q5.3

1 Point

What would be an appropriate conclusion?

We reject the null hypothesis. We found evidence of a linear relationship between lifespan and weight for dog breeds.

We fail to reject the null hypothesis. We found evidence of a linear relationship between lifespan and weight for dog breeds.

We reject the null hypothesis. We did not find evidence of a linear relationship between lifespan and weight for dog breeds.

We failed to reject the null hypothesis. We did not find evidence of a linear relationship between lifespan and weight for dog breeds.

Save Answer

## Q5.4

1 Point

We found evidence of a linear relationship, but let's provide an estimate with uncertainty of how much we expect the lifespan to differ for a one pound increase in weight of dog breeds.

```
> confint(dog_model)
              2.5 %      97.5 %
(Intercept) 12.75297938 14.01792347
weight_lb   -0.05209167 -0.03236664
```

What is the 95% confidence interval for the slope of the regression? Round to four decimal places.

Lower bound:

Upper bound:

Save Answer

**Q6****3 Points**

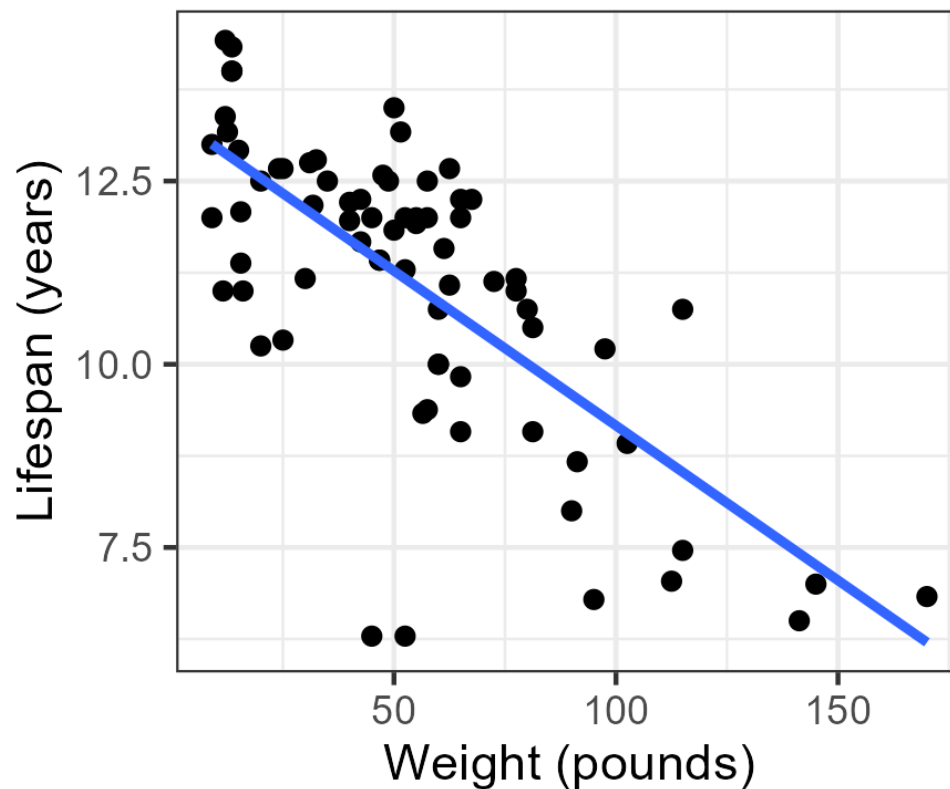
Two key things that linear regression is helpful for are:

- Investigating the expected difference in the response variable when the explanatory variable changes (through the estimate and confidence interval of  $\beta_1$ ). *This is what we did in question 5.*
- Predicting the value for the response variable, given a value for the explanatory variable. *This is what we will do in question 6.*

## Q6.1

1 Point

The linear regression line we fit is plotted below over the data.



Specify our estimated equation for the regression line (hint: look at the output shown in question 5.2):

[Save Answer](#)



**Q6.2****1 Point**

Bearded Collies have an average weight of 50 pounds. What would we estimate their lifespan to be (in years)? Enter your numeric answered and upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q6.3****1 Point**

Increasing a dog's weight will decrease their lifespan.

True

False

We can't say from this analysis

**Save Answer**

**Q7****1 Point**

The following two questions are related to the homework choice you made in choosing to do this assignment instead of the other two options. There are no correct answers, this question is only being graded for completeness (note that answering "I felt like it" or "it was interesting" is an insufficient response).

Please explain in detail what led you to choose this topic, dog lifespan, to investigate.

What was it about the other two options (teen pregnancy and typing speed) that led you not to choose them? Please explain in detail.

[Save Answer](#)

**Q8****0 Points**

- Have you done the homework by yourself, with a peer(s), or used tools like ChatGPT? If you do not acknowledge any help from a person or ChatGPT, we will consider this cheating.

- If you provided help to a peer(s) and/or got help from your peers(s) please explain briefly what this entailed.

- If you used ChatGPT, include the prompts you used and explain how ChatGPT supported your learning.

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